Proceedings of CONF-MLA 2025 Symposium: Intelligent Systems and Automation: AI Models, IoT, and Robotic Algorithms
DOI: 10.54254/2755-2721/2025.LD29658

A Review of Generative Models for Training Datasets of Scene Text Recognition

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Abstract. The information extracted by Scene Text Recognition (STR) is of significance to various vision-based application due to the rich semantic information contained in the text within images. However, the development of STR is hindered by challenges arising from variations in font, size, and distortion, among others. This paper focuses on multiple techniques for generating the training datasets of STR, comparing and analyzing their effects and working principles. Employing a literature analysis and review methodology, this paper aims to construct an overview of the field by categorizing the techniques into rendering-based, GAN-based, Diffusion-based, and other methods. Furthermore, it provides recommendations for the future development of synthetic datasets. This paper finds that mainstream STR training datasets generation techniques inherently involve a trade-off between image realism and text controllability. Therefore, the core challenge in the current field is to establish a unified evaluation criterion to bridge the gap between synthetic and real-world data.

Keywords: Scene Text Recognition, Scene Text Synthesis, Optical Character Recognition

1. Introduction

Throughout history, from ancient oracle bone script to modern digital encoding, text has served as the main carrier of information. Text manifests in diverse forms across daily life, such as on road signs, in graffiti, and in documents. These physical instances are frequently captured and recorded as images. The text information contained in the images offers rich semantic information, which is of great significance to various vision-based applications, comprising robot control, autonomous vehicle navigation, among others. Specifically, Scene Text Recognition (STR), as a branch of Optical Character Recognition (OCR), is dedicated to extracting text from natural and real-world scenes. Compared with OCR, STR is confronted with images that have more complex backgrounds, more diverse text forms, higher sensitivity to noise, and whose texts are captured randomly. Consequently, the tasks of STR are more challenging [1].

Training a STR model requires appropriate training datasets, specifically scene text images, and synthetic datasets are currently widely adopted. Though Baek et al. [2] suggested that STR models achieve superior performance when trained exclusively on real-world datasets, Luo et al. [3] argued that the high cost and limited coverage of types in creating real-world datasets necessitate the use of synthetic datasets. Furthermore, a comprehensive survey specifically dedicated to the techniques for

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DOI: 10.54254/2755-2721/2025.LD29658

generating synthetic STR datasets is currently lacking in the literature, which thus motivates this paper.

The remainder of this paper is organized as follows: Section 2 introduces the background and the basic principles of various scene text image synthesis techniques. Section 3 enumerates some current models utilizing these techniques. Section 4 concludes the paper, and identifies the potential direction for future work in synthetic STR training datasets generation. Therefore, the main contribution of this paper is summarized as follows: Categorizing the methods of synthesizing scene text images based on their underlying techniques; Providing advice for the future development of synthetic datasets.

2. Background

2.1. Rendering-based scene text image synthesis

The rendering-based methods are the most traditional and straightforward techniques of scene text image synthesis, which offer the merits of high speed, low cost and strong controllability [4]. However, a notable domain gap exists between the synthesized images and real-world data [3]. The core principle of this type of technique, as summarized from some of the existing literature, is to simulate the rendering process in computer graphics. This involves programmatically mapping text characters onto a background image or 3D scene, followed by applying a series of transformations and blending techniques to mimic real-world text appearance [4,5]. Taking SynthTIGER as an example, the generation pipeline typically comprises: text shape selection, text style selection, transformation, blending, and post-processing [6].

2.2. GAN-based scene text image synthesis

In contrast, Generative Adversarial Network (GAN)-based methods represent a further advancement over rendering-based techniques, aiming to enhance the realism of synthesized images while simultaneously reducing the domain gap. However, these methods, especially earlier architectures, suffer from unstable training and are prone to mode collapse [7], and they often lack precise control over the textual or glyph structure [8]. The core principle of this type of technique involves introducing an adversarial learning mechanism, comprising a Generator and a Discriminator. In this process, both components receive the text label as an auxiliary condition. The Generator then generates an image for the Discriminator to distinguish from real samples. They are trained adversarially and promote each other's capability, finally enabling Generator to produce synthetic scene text images that highly resemble real-world images [9].

2.3. Diffusion-based scene text image synthesis

The most cutting-edge and effective methods in the current image generation field are the Diffusion-based techniques. Its generated images surpass those of GANs in terms of realism [10], while it still suffers from the risk of generating spelling errors [11]. The core principle of this type of technique is a probabilistic generation process inspired by thermodynamics, comprising a forward process and a reverse process. The former one is a fixed, non-learned process that model adds noise to the images gradually and constantly until it becomes a pure noise distribution; The latter one is conditioned by the text encoding, allowing the model to progressively predict the noise added at each step and eliminate it. This guidance enables the generated images to converge on the desired text content and structure [12]. Additionally, a key difference is that the input for Diffusion-based methods is a

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descriptive text prompt, contrasting with the required input of the two previous techniques (the target text and the raw background image).

2.4. Other methods

Beyond the three major techniques mentioned above, other effective approaches also exist, which are typically hybrid or image-editing-based methods. Take image-editing-based methods as an example, they can maximally preserve the authentic lighting, shadows, and background of the original image. However, a drawback is that the generated text's position and style are constrained by the source image. Its core principle involves erasing the original text from a real image, inpainting the background texture to restore the clean background, and finally using a rendering-based method to synthesize the new target text into the original text's location for replacement [13]. In addition, there are emerging methods, such as those based on Neural Radiance Fields (NeRF), which will be discussed in Section 3.

3. Methods review

3.1. Rendering-based models

The earliest model utilizing this technique dates back to MJSynth [14], proposed by Jaderberg et al. in 2014, while it only generated images with non-realistic backgrounds and single words. Later, Gupta et al. introduced SynthText [4], which generated images with real-world background and multiple words successfully. Subsequently, the Verisimilar Image [15] proposed by Zhan et al. addressed the limitation of SynthText, where text could be rendered onto inappropriate surfaces, by employing selective semantic segmentation. Afterward, the SynthTIGER [6] proposed by Yim et al. combines the advantages of MJSynth (MJ) and SynthText (ST) while alleviating the long-tailed recognition.

Additionally, this type of technique includes another category of approaches, which utilize 3D engines for image generation. These examples include the SynthText3D [5] proposed by Liao et al., and the UnrealText [16] proposed by Long et al., among others. Both examples above utilize threaded 3D engines (such as UE4) for scene text synthesis. However, the former (SynthText3D) focuses more on exploiting the engine's high-fidelity capabilities to render the text and the virtual scene as a unified whole, thereby achieving more complex perspective transforms, various illuminations, and occlusions. The latter (UnrealText), in contrast, emphasizes an efficient, procedural workflow to control and generate large-scale, diverse datasets, better bridging the domain gap between synthetic and real-world data.

3.2. GAN-based models

Following the pioneering work on text-to-image generation, StackGAN [17], proposed by Zhang et al., GAN-based text-to-image synthesis models gradually evolved. It was not until 2019 that Fang et al. introduced STS-GAN [8], making the formal entry of GANs into the field of scene text synthesis. It synthesizes realistic text images by treating text as a specific type of texture, allowing it to render text with various styles and appearances onto backgrounds. The model uses a multi-scale discriminator to ensure the visual quality of the generated text. In the same year, Zhan et al. proposed SF-GAN [18], which combined a geometry synthesizer and an appearance synthesizer to achieve photo-realistic synthesis in both geometric and appearance spaces, and applied this to scene text image synthesis for training recognition models. Subsequently, in 2020, Kang et al. introduced

DOI: 10.54254/2755-2721/2025.LD29658

GANwriting [19], the first model to use GANs for content-conditional handwritten word image generation, capable of mimicking the writing style of a reference sample and synthesizing realistic handwritten text of arbitrary content. However, the performance of GAN models in generating from scratch has gradually been surpassed by Diffusion models. Therefore, its research focus shifted to the STE task of modification based on real images. Subsequently, GAN-based models like SwapText [20] and TextStyleBrush [21] emerged.

3.3. Diffusion-based models

Since Nichol et al. proposed GLIDE [22] in 2021, text-guided diffusion models for image generation have gradually gained widespread attention. The following year, Ramesh et al. introduced DALL-E2 [23], which demonstrated impressive general image generation, while the embedded text often suffers from spelling errors, that is, it could not generate images with text stably. By 2023, Chen et al. introduced TextDiffuser [24], addressing the core limitation of models such as DALL-E2 and achieving significant advancements in the stability, accuracy and coherence of text rendering. Also in 2023, Ma et al. introduced Glyphdraw [25], which successfully generates images with embedded Chinese text, but its accuracy decreases with an increased character count, is restricted to short text, and lacks support for multiple bounding boxes. Later that year, Wang et al. introduced AnyText [26], which improved the rendering quality of complex scripts like Chinese by incorporating a Character-aware Supervision mechanism. This significantly enhanced text fidelity and also enabled multilingual generation and text editing. In 2024, “Brush your text” [27] introduced by Zhang et al., designed a scene text generation framework that requires no training and supports more languages, Diff-Text, which demonstrated strong performance in generating scene text images for various languages, including Chinese, English, Russian, Thai, and Arab, among others.

3.4. Other models

3.4.1. Scene text editing models

This section will briefly review other methods, particularly those that synthesize new scene text images via STE. In 2019, the Style Retention Network (SRNet) [13] introduced by Wu et al. decomposed the editing task into three modules: a text conversion module, a background inpainting module, and a fusion module. The work claims to be the first to address the problem of word or text-line level STE by an end-to-end trainable network. The following year, Roy et al. proposed STEFANN [28], focusing on achieving more refined character-level editing. Also in 2020, Yang et al. introduced SwapText [20], which refined the text replacement workflow by incorporating a spatial transformation module (such as TPS) to better handle arbitrary or curved text. In recent years, the rapid development of diffusion models has led to the emergence of diffusion-based approaches for STE. For instance, in 2023, Ji et al. proposed DiffSTE [29], which innovatively introduced a dual-encoder design, significantly boosting editing accuracy and image realism. In the same year, Chen et al. proposed DiffUTE [30], which provided a unified framework to handle various text editing tasks. In 2024, the TextCtrl [31] proposed by Zeng et al. further enhanced scene editing capability by introducing prior guiding control. In 2025, GlyMastero [32] proposed by Wang et al. achieved even more precise control at the stroke-level. While maintaining a comparable Fréchet Inception Distance (FID) to TextCtrl’s generated images, it significantly boosted the editing accuracy, resulting in synthesized images that rival real ones.

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3.4.2. Novel synthesis paradigms

As for alternative methods beyond STE, only a few examples are provided here. For instance, in 2023, Tang et al. proposed the LBTS [33] engine, which is capable of learning from real-world data to synthesize scene text images that are more photo-realistic and diverse than those produced by traditional methods. In 2024, Cui et al. pioneered the use of Neural Radiance Field (NeRF) technology for scene text synthesis [34], and simultaneously proposed the first data set to include 3D pose attributes.

4. Conclusion

This study categorized the techniques for generating Scene Text Recognition (STR) training datasets into rendering-based, GAN-based, diffusion-based, and other methods. It introduced the principles of each technique and chronologically reviewed some of the models utilizing these various approaches. However, despite the introduction stating that using real data yields better performance for STR models, this review lacks a unified standard for comparing the quality gap between the generated images and real images, so it is difficult to judge the superiority or inferiority of the model by combining all aspects of factors. The future research direction for STR data set synthesis work will be to establish a set of evaluation criteria to measure the domain gap between real-world and synthetic data, and to subsequently explore methods to narrow this gap. Moreover, future evaluation criteria should not only include metrics for measuring image realism (such as FID and KID), but also specialized metrics for assessing text rendering quality (e.g., Character Error Rate (CER and glyph similarity), and most importantly, 'task-oriented' metrics that quantify their improvement in downstream STR tasks.

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